

Muhammad Shayan

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EDUCATION AND CERTIFICATIONS

Coventry University, **BSc Computer Science with Artificial Intelligence** September 2023 - July 2026
modules: Artificial Intelligence, Machine Learning, Artificial Neural Networks, Data Science, Software Engineering, Security, Robotics & Intelligent Agents, Operating Systems, Theory of Computation

Coursera: **IBM Data Science Professional Certificate** August 2024 - January 2025

Modules included: Databases and SQL, Data Science Methodology, Data Visualization, Data Analysis, Generative AI

Coursera: **Extended Reality for Everybody Specialisation** February 2025 - April 2025

Modules Included: Development and Introduction to AR/VR/MR/XR Apps with WebXR, Unity & Unreal

WORK EXPERIENCE

Course Representative Coventry University - Student Union March 2024 - April 2026

- Maintained ongoing stakeholder relationships across a cohort of 300 to 350 students over two years, consolidating recurring concerns into points and presenting them to module leaders in course meetings, tracking each issue till confirmed resolutions.
- Identified widespread concern that the Final Year Project word limit of 6,000 words was too restrictive for the technical depth the project required. Brought a formal case to the department and secured the addition of appendices, code listings, and figures as official exclusions, resulting in a meaningful increase to the effective submission scope for the entire cohort.

PROJECT EXPERIENCE

Better AI Image Recognition & Translation - Final Year Project November 2024 - April 2025

Python, OpenCV, Tesseract, EasyOCR, Keras-OCR, SVM, Google Translate API

- Developed an OCR pipeline targeting manga, handwritten, and stylised text, image types for which standard tools consistently produce poor results for due to irregular fonts and overlaid graphics. Tesseract was used as the OCR, with a method selecting the best result across 31 preprocessing runs, improving accuracy from 71% on a single pipeline to 80% on unseen test images.
- Trained a Support Vector Machine classifier on a labelled dataset of 56 images across the three categories, achieving 92.9% accuracy, enabling automatic routing of each input image to its appropriate preprocessing pipeline without manual intervention. A multi-language translation layer and an interactive interface built with ipywidgets were included, producing a fully self-contained tool operable without any modification to the underlying code.

KomodoHub - Multi-User Educational Platform (Group Project) March 2025

Python Flask, SQLite, HTML/CSS/JavaScript

- Worked across all 3 agile roles over 2 sprints, rotating weekly between Scrum Master, Product Owner, and Developer, facilitating sprint planning sessions, maintaining the product backlog, coordinating task allocation, and intervening directly in teammates' code to diagnose and resolve integration issues that were blocking delivery.
- Deployed core backend responsibilities including session authentication with password hashing, Base64 image storage, and CORS configuration to support platform features such as community and student posts with image uploads, user profiles with point tracking, proper database connectivity and shared content libraries.
- Designed and built a Wordscapes-inspired game independently across 2 sprint cycles, consisting of 3 difficulty levels, drag-to-select letter mechanics, a difficulty-scaled hint and reveal system, and database-linked score tracking

Social Bias Detection - Neural Network Coursework March 2026

Python, TensorFlow/Keras, Pandas, Scikit-learn

- Trained multi-output neural networks on 144,649 social media posts to classify offensive, intent, and sexual content. Focal loss was used to handle class imbalance and reduce overconfidence on harder cases, with annotator demographic skew flagged as a fairness concern before deployment. The result was an F1 score of 0.85 on the primary offensive class.
- Exploratory data analysis on the same dataset informed every architectural decision, including the selection of a 1D CNN architecture over a bag-of-words approach, following token frequency analysis that confirmed word order and contextual structure, rather than vocabulary alone, as the primary distinguishing signal between classes.

Secure E-Commerce Backend - Security Module November 2025

Python Flask, SQLite, Flask-WTF, Flask-Limiter

- Designed a multi-role e-commerce backend for customers, sellers, and administrators, with security built in at every layer rather than added as an afterthought. Protections included CSRF via Flask-WTF, Content Security Policy headers to prevent XSS, clickjacking prevention, rate limiting on sensitive endpoints, HTTPOnly session cookies, and strict file upload validation. Bandit static analysis and safety dependency audits were run and submitted as part of the deliverable, demonstrating that the system had been tested against known vulnerabilities.
- Prior to implementation, each component was threat modelled against the OWASP Top 10 to systematically map specific risks to their corresponding controls, CSRF on transaction endpoints, stored XSS on product reviews, and IDOR on order histories, ensuring every protection addressed a threat and that no attack surface was left unmitigated.

TECHNICAL SKILLS

Languages: Python, JavaScript, C++, HTML/CSS, SQL, Bash **Frameworks:** Flask, React, Node.js

Data & ML: TensorFlow/Keras, Scikit-learn, Pandas, Matplotlib, OpenCV **Tools:** Git, REST APIs, MongoDB, SQLite, Docker

Security: CSRF/XSS/CSP, session hardening, bandit, Flask-WTF, Flask-Limiter **Exploring:** TypeScript, GraphQL